

Static Routing

What is Static Routing?

- A routing method where **routes are manually configured** by an administrator on routers.
- The router **does not learn or adjust** routes automatically; it forwards packets strictly per the configured routes.
- Used to define **fixed, deterministic paths** between networks.

Key Characteristics

- **Manual configuration:** Admin enters destination network, subnet mask, and next-hop (or exit interface).
- **No routing protocol** overhead (no hello/LSA/updates).
- **Predictable** behavior; routing table changes only when admin modifies it.
- **Resource-light:** minimal CPU/RAM and bandwidth usage.

Advantages

- **Simplicity** for small/flat networks.
- **Low overhead** (no CPU/bandwidth spent on routing updates).
- **High security/control:** no unexpected path changes.
- **Deterministic:** good for **stub networks**, DMZs, static default routes, or backup paths.

Disadvantages

- **No automatic failover:** if a link goes down, traffic is black-holed until admin intervenes (unless paired with tracking).
- **Poor scalability:** becomes **labor-intensive** in medium/large topologies.
- **Human error risk** in manual entries (wrong mask/next-hop).
- **No metric optimization:** cannot auto-choose best path.

Common Use Cases

- **Small branch/stub networks** (single exit to HQ/Internet).
- **Default route to ISP** on edge routers.
- **Transit/point-to-point links** with stable paths.
- **Backup routes** (floating statics) behind a dynamic protocol.
- **Routing to static services** (e.g., management networks, loopbacks).

Core Concepts

- **Next-Hop vs Exit Interface:**
 - Next-Hop IP (e.g., 10.0.0.2) tells the router where to send the packet.
 - Exit interface (e.g., G0/0) specifies the outgoing port (common on point-to-point links).
- **Default Route:** a catch-all for unknown destinations (0.0.0.0/0).
- **Floating Static:** a static route with **higher administrative distance** than a dynamic protocol, used as an **automatic backup**.
- **Administrative Distance (AD):** trust value; lower is preferred (Static AD = 1; floating static uses higher AD like 5–254).

Verification & Troubleshooting

- **Show routes:** `show ip route` (look for “S” entries for Static).
- **Test reachability:** `ping <ip>`, `traceroute <ip>`.
- **Check neighbors/ARP:** `show arp`, `show ip cef <ip>`.
- **Packet path:** `debug ip packet` (use with caution), or `show ip route <prefix>`.

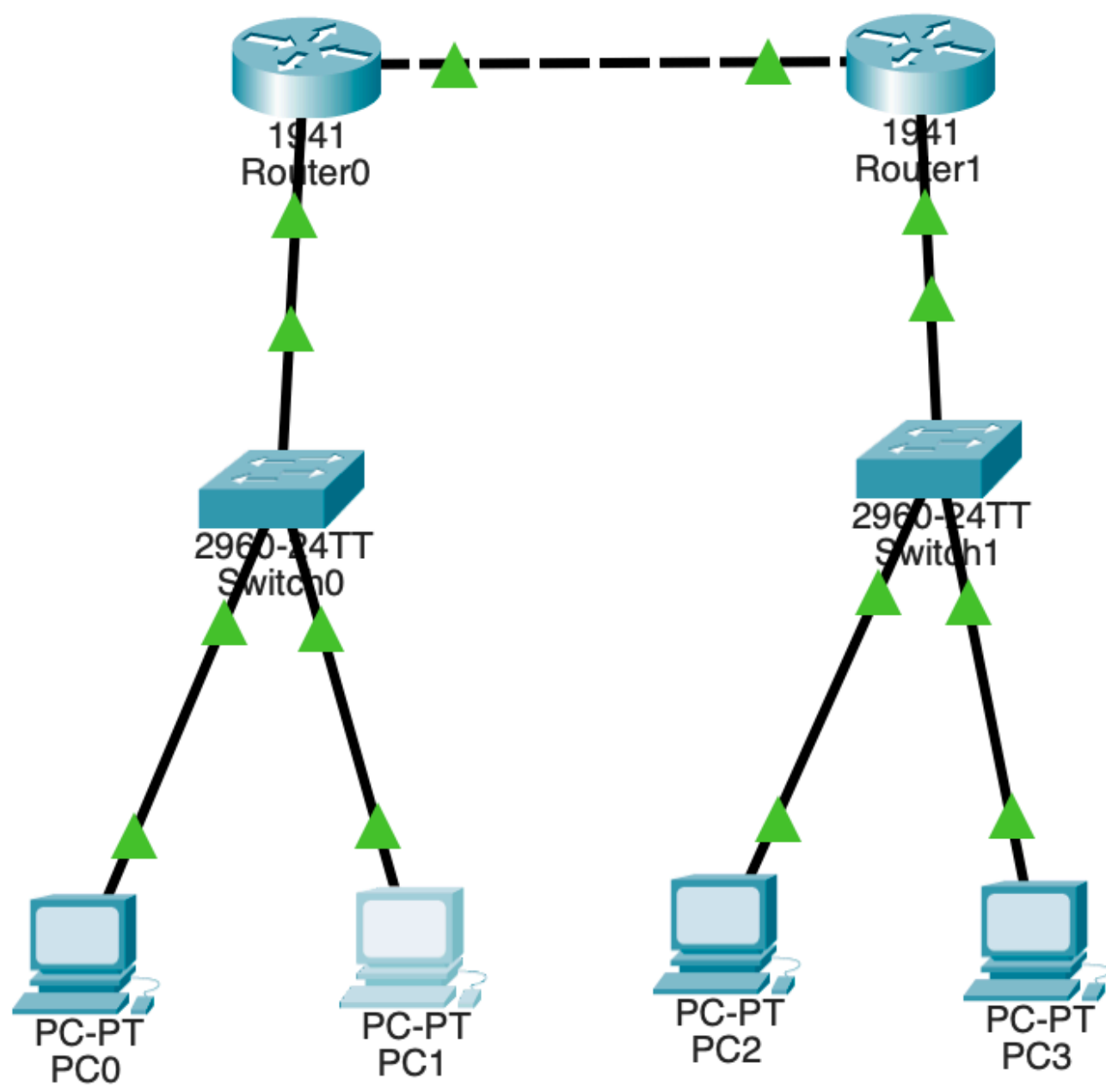
Best Practices

- Keep a **clean IP plan** (subnets, masks, next-hops documented).
 - Use **default + a few specifics** to minimize entries.
 - For resilience, deploy **floating statics** as backups to dynamic routing.
 - Avoid recursive next-hop loops; ensure next-hop is reachable.
 - On multiaccess networks, prefer **next-hop IP**; on p2p links, **exit interface** is fine.
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1. Devices Required

- 2 Routers (e.g., Cisco 1941)
- 2 Switches (e.g., 2960-24TT)
- 4 PCs
- Copper Straight-Through Cables (for PC ↔ Switch and Switch ↔ Router)

- Copper Cross-Over Cable (for Router ↔ Router link)



2. Network Topology

Left Side (Router0 LAN) → PC0, PC1 → Switch0 → Router0

Right Side (Router1 LAN) → PC2, PC3 → Switch1 → Router1

Then connect **Router0** ↔ **Router1** directly using **GigabitEthernet0/1** ports.

3. IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	GigabitEthernet0/1	10.0.0.1	255.255.255.252	—
Router1	GigabitEthernet0/0	192.168.2.1	255.255.255.0	—
Router1	GigabitEthernet0/1	10.0.0.2	255.255.255.252	—
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

4. Step-by-Step Configuration

Step 1 – Connect Devices

1. Connect PCs to switches using **Copper Straight-Through** cables.
2. Connect each switch to its respective router's **GigabitEthernet0/0** port.
3. Connect **Router0** → **Router1** using **GigabitEthernet0/1** (Cross-Over cable).

Step 2 – Configure Router0 Interfaces

1. Click **Router0** → **Config tab** → **Interface** → **GigabitEthernet0/0**
 - IP Address: **192.168.1.1**
 - Subnet Mask: **255.255.255.0**
 - Click **ON**.
2. Select **GigabitEthernet0/1**
 - IP Address: **10.0.0.1**
 - Subnet Mask: **255.255.255.252**
 - Click **ON**.

Step 3 – Configure Router1 Interfaces

1. Click **Router1** → **Config tab** → **Interface** → **GigabitEthernet0/0**
 - IP Address: **192.168.2.1**
 - Subnet Mask: **255.255.255.0**
 - Click **ON**.
2. Select **GigabitEthernet0/1**
 - IP Address: **10.0.0.2**
 - Subnet Mask: **255.255.255.252**
 - Click **ON**.

Step 4 – Configure Static Routing

On Router0:

Go to → **Config** → **Routing** → **Static**

Fill in:

Field	Value
Network	192.168.2.0
Mask	255.255.255.0
Next Hop	10.0.0.2

Click **Add**

On Router1:

Go to → **Config** → **Routing** → **Static**

Fill in:

Field	Value
Network	192.168.1.0
Mask	255.255.255.0
Next Hop	10.0.0.1

Click **Add**

Step 5 – Configure PCs

Click each PC → **Desktop** → **IP Configuration**

Left Side (Router0 LAN)

PC	IP	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1

Right Side (Router1 LAN)

PC	IP	Subnet Mask	Default Gateway
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

5. Verification

Check interface status:

On each router → CLI tab:

```
show ip interface brief
```

All should show “**up/up**”.

Check routing table:

```
Router1> exit
Router1> enable
Router1# show ip route
```

```
show ip route
```

- On Router0 → You should see: S 192.168.2.0 [1/0] via 10.0.0.2
 - On Router1 → You should see: S 192.168.1.0 [1/0] via 10.0.0.1
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Ping Test:

From **PC0**, open Command Prompt:

```
ping 192.168.2.2
```

Successful reply = Static Routing working!
